# ASSIGNMENT 4

## 4.Take one domain and build business understanding.

## Domain: Banking Sector

**Project Title**: Credit Card Fraud Detection System

## 1. Business Understanding

### Problem Statement

- Banks provide credit card services to millions of customers. Some transactions are fraudulent (unauthorized usage), which causes financial loss to banks and customers. Detecting fraud late increases monetary loss and reduces customer trust.
- Therefore, the bank needs a system to identify fraudulent transactions in real-time.

### Business Objectives

- Detect fraudulent credit card transactions.
- Minimize financial losses.
- Reduce false declines of genuine transactions.
- Improve customer trust and satisfaction.
- Strengthen bank security systems.

### Preventive Actions:

- Real-time transaction blocking.
- OTP verification for suspicious transactions.
- Temporary card suspension.
- Alert notifications to customers.

### Stakeholders

- Bank customers
- Fraud detection team
- Risk management department
- IT & Security team
- Data science team
- Bank management

### Business Success Criteria

- Reduce fraud loss by 20–30%.
- Increase fraud detection rate.
- Reduce false positives.
- Improve customer satisfaction score.
- Ensure compliance with financial regulations.

## 2.Assess Situation

**Inventory of Resources**

- Historical transaction data
- Customer account details
- Transaction amount and time data
- Merchant information
- Fraud-labelled dataset
- Banking security team
- IT infrastructure (servers, cloud, databases)

**Requirements**

- High fraud detection accuracy.
- Real-time prediction capability.
- Low false positive rate.
- Explainable model for regulatory compliance.

**Assumptions**

- Historical fraud patterns can predict future fraud.
- Data provided is clean and accurate.
- Fraud behaviour patterns remain somewhat consistent.
- Customers use cards regularly.

**Constraints**

- Highly imbalanced dataset (fraud cases are rare).
- Data privacy and banking regulations.
- Need for real-time processing.
- Limited labelled fraud data.

**Risks**

- False positives may block genuine customers.
- False negatives may cause financial loss.
- Cybercriminal behaviour may change.
- Model performance may degrade over time.

**Costs and Benefits**

**Costs:**

- Infrastructure setup.
- Model development and maintenance.
- Skilled manpower.
- Security compliance cost.

**Benefits:**

- Reduced financial fraud loss.
- Increased customer trust.
- Improved brand reputation.
- Better regulatory compliance.

## 3. Data Science Goals

**Data Science Objective**

- Build a binary classification model to predict:
- Fraudulent Transaction (Yes)
- Genuine Transaction (No)

**Data Science Tasks**

- Data preprocessing and cleaning.
- Feature engineering (transaction frequency, time gaps, location mismatch).
- Handling class imbalance (SMOTE / under sampling).
- Model building using:
    - Logistic Regression
    - Decision Tree
    - Random Forest
    - XGBoost
- Model evaluation and hyperparameter tuning.

**Data Science Success Criteria**

- Accuracy > 90%
- Recall (Fraud class) > 85%
- Precision > 80%
- AUC-ROC > 0.85
- Low False Positive Rate

## 4. Project Plan

| Stage | Activity | Duration |
|---|---|---|
| 1 | Business Understanding | 1 week |
| 2 | Data Collection & Understanding | 2 weeks |
| 3 | Data Cleaning &Preparation | 2 weeks |
| 4 | Model Building | 3 weeks |
| 5 | Model Evaluation | 1 week |
| 6 | Deployment | 1 week |
| 7 | Monitoring &Maintenance | Continuous |

**Resources Needed**

- Data Scientists
- Banking domain experts
- IT & Security engineers
- Risk management team

**Tools & Technologies**

- Python (Pandas, NumPy, Scikit-learn)
- SQL
- Power BI / Tableau
- Machine Learning Algorithms
- Cloud Deployment (AWS / Azure)

## Final Outcome

- A deployed Fraud Detection System.
- Real-time fraud alerts.
- Reduced financial loss.
- Increased security and customer trust.
- Alignment of business and data science objectives.